CSE422 Quiz: Informed Search — Solution Key
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Scenario: You are solving a maze on a 5×5 grid with Start $S(x_n, y_n)$ and Goal $G(x_G, y_G)$. Moves: up, down, left, right, and diagonal. One straight move cost = 1, one diagonal move cost = $\sqrt{2}$.

$$h_O(n) = (\sqrt{2}-1)\min(dx, dy) + \max(dx, dy), \qquad dx = |x_n - x_G|,\ dy = |y_n - y_G|.$$

Part 1: Admissibility of h_O

Question: Considering the following cases, compute $h_O(n)$ and check admissibility.

a. [2 points.] **Case 1:** $dx = dy$**.** Considering no obstacles, the minimum distance from n to G is $\sqrt{2}dx = \sqrt{2}dy$.

> Solution
>
> $$h_O(n) = (\sqrt{2}-1)\min(dx, dy) + \max(dx, dy) = (\sqrt{2}-1)dx + dx = \sqrt{2}dx.$$
>
> With no obstacles, true path cost = $\sqrt{2}dx$. Hence $h_O(n)$ = true cost (exact) $\Rightarrow$ **admissible**.

b. [2 points.] **Case 2:** $dx > dy$**.** Considering no obstacles, the minimum distance from n to G is $\sqrt{2}dy + (dx - dy)$.

> Solution
>
> $$h_O(n) = (\sqrt{2}-1)dy + dx = dy\sqrt{2} + (dx - dy).$$
>
> With no obstacles, true cost = $dy\sqrt{2} + (dx - dy)$. Thus $h_O(n)$ = true cost (exact) $\Rightarrow$ **admissible**.

Part 2: Straight Movement Consistency Check

Suppose the agent moves one step **along** x toward the goal: $dx' = dx - 1$, $dy' = dy$, and $c(n, n') = 1$. Check that $h_O(n) \le 1 + h_O(n')$. **Note:** Computations for straight movement along the y direction are similar, with x replaced by y.

(a) [2 points.] **Case 1:** $dx = dy$**.**

> Solution
>
> $$h_O(n) = \sqrt{2}dx.$$
>
> After moving: $dx' = dx - 1 < dy' = dx$, so
>
> $$h_O(n') = (\sqrt{2}-1)\min(dx', dy') + \max(dx', dy') = (\sqrt{2}-1)(dx - 1) + dx.$$
>
> $$h_O(n) - h_O(n') = \sqrt{2} - 1 < 1 \Rightarrow h_O(n) \le 1 + h_O(n') \checkmark$$

(b) [4 points.] **Case 2:** $dx > dy$. A straight step along x may lead to $dx - 1 > dy$ or $dx - 1 = dy$.

Solution

(i) $dx - 1 > dy$: $\min(dx', dy') = dy, \ \max(dx', dy') = dx - 1$

$$h_O(n') = (\sqrt{2} - 1)dy + (dx - 1) = h_O(n) - 1 \Rightarrow h_O(n) = 1 + h_O(n') \checkmark$$

(ii) $dx - 1 = dy$: $dx' = dy' = dy$

$$h_O(n') = \sqrt{2}dy, \quad h_O(n) = (\sqrt{2} - 1)dy + (dy + 1) = \sqrt{2}dy + 1.$$

$$h_O(n) = 1 + h_O(n') \checkmark$$

Part 3: Diagonal Movement Consistency Check

A diagonal move toward the goal reduces both distances by 1: $dx' = dx - 1$, $dy' = dy - 1$, with move cost $c(n, n') = \sqrt{2}$. Check:

$$h_O(n) \leq \sqrt{2} + h_O(n').$$

(a) [2 points.] **Case 1:** $dx = dy$**.**

Solution

$$h_O(n) = \sqrt{2}dx, \quad h_O(n') = \sqrt{2}(dx - 1).$$

$$h_O(n) - h_O(n') = \sqrt{2} = c(n, n') \Rightarrow h_O(n) = \sqrt{2} + h_O(n') \checkmark$$

(b) [2 points.] **Case 2:** $dx > dy$**.**

Solution

$$h_O(n) = (\sqrt{2} - 1)dy + dx, \quad h_O(n') = (\sqrt{2} - 1)(dy - 1) + (dx - 1).$$

$$h_O(n) - h_O(n') = \sqrt{2} = c(n, n') \Rightarrow h_O(n) = \sqrt{2} + h_O(n') \checkmark$$

(4) [1 point.] **Case 3:** $dy > dx$**.**

Solution

This case mirrors Case 2 ($dx > dy$). Swap dx and dy in the formulas. Result: $h_O(n) - h_O(n') = \sqrt{2} = c(n, n')$. Hence, $h_O(n) = \sqrt{2} + h_O(n') \checkmark$

Conclusion.

For an 8-connected grid with straight cost 1 and diagonal cost $\sqrt{2}$:

- h_O is **admissible** — it never overestimates the true cost (exact when no obstacles).
- h_O is **consistent** — for every straight or diagonal step,

$$h_O(n) \leq c(n, n') + h_O(n'),$$

and equality holds for goal-directed moves.